

QSS 20: Take-Home 1

Modern Statistical Computing | Dartmouth College | Spring 2026

Logistics

Format: Individual, open-note, open-documentation. You may use course materials, Python documentation, and any resources from class. **You may not collaborate with classmates.**

Submission: Submit a Jupyter notebook (.ipynb) containing your code for Parts 1-3, plus a separate 1-page interpretation writeup (as a PDF) for Part 4. Upload both to Canvas by Sunday 11:59 PM EST.

Timeline: Due Sunday night. You should not need more than a few hours of focused work.

Context

You are working with the Cook County State Attorney's Office felony sentencing dataset, which contains records of felony cases processed in Cook County, Illinois. Each row represents a charge within a case. Defendants may have multiple charges per case.

Your task is to inspect, clean, and analyze this data to investigate a simple question:

"Are there racial disparities in incarceration rates?"

Structure

The take-home has four parts. Parts 1-3 are completed in the Jupyter notebook provided. Part 4 is a written interpretation submitted as a separate PDF.

Parts 1-3: Jupyter Notebook (65 points)

The notebook contains all coding questions. The three sections are:

Part 1: Data Inspection & Understanding (15 points) — Examine the dataset's dimensions, data types, and unit of analysis.

Part 2: Data Cleaning (20 points) — Recode the RACE variable and subset to primary charges.

Part 3: Descriptive Analysis (30 points) — Compute incarceration rates by race and create a visualization.

Part 4: Written Interpretation (35 points)

Submit a separate 1-page PDF (no more than one page) that addresses the following three questions.

4a. What do your results suggest? (15 points)

In 3-4 sentences, describe what your analysis from Part 3 reveals about racial disparities in incarceration rates. Be specific and reference the actual numbers or patterns you found.

4b. What could be confounding your results? (10 points)

Name one specific variable in this dataset that could partly explain the gap you found. Explain in 1-2 sentences why it might be a confounder — i.e., how it could be correlated with both race and incarceration.

4c. What would you do next? (10 points)

If you wanted to investigate whether the racial gap is driven by differences in the types of crimes different groups are charged with, describe in a short paragraph what steps you would take. No code required — just describe your approach that a colleague could follow it.

Grading

Component	Points
Part 1: Data Inspection & Understanding	15
Part 2: Data Cleaning	20
Part 3: Descriptive Analysis	30
Part 4: Written Interpretation	35
Total	100

Note: For Parts 1-3, you will be graded on both correctness and code clarity. Use comments where helpful. For Part 4, you will be graded on the quality of your reasoning, not whether you reach a particular conclusion.

Tips

- You do not need to analyze every column. Focus on what the questions ask.
- If you get stuck on one part, move on. The parts are mostly independent.
- For Part 4, write for a reader who hasn't seen your code. Be concrete.

AI Use Reflection

After your one-page memo, also answer the following questions:

- Did you use generative AI?
- If so, what model did you use?
- If you didn't use it, why not?
- Describe your workflow in a paragraph.
- Did you run out of tokens? What were your biggest constraints?

Remember to attach your transcripts, if you'd like. While this is optional, I reserve the right to do brief oral check-ins for both content that is not disclosed and disclosed content to check your understanding. Depending on your model, you can either just print your chat or export it directly.